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9.1 Poisoning

General supportive treatment using other guidelines should be implemented for all poisonings. This guideline provides further procedures for the care of poisoned or chemically exposed patients.

NOTICE **The Poison Control Center should be contacted for assistance in managing the patient at 800–222–1222**

- Identify the type and amount of substance the patient was exposed to
- Establish the time of exposure
- If available, medication containers or chemical labels should be photographed, or taken to the hospital with the patient if safe

9.1.1 Opioid Overdose

Administer naloxone for opioid induced life threatening respiratory depression or apnea

Naloxone

Adult

- 1-2 mg IV/IM/IN

NOTE: **Full reversal of opioid intoxication/overdose**

Pediatric

- 0.01 mg/kg IV/IM/IN
- Max single dose 1 mg

Naloxone—Infusion

Adult

- Determine the initial dose required to reinstate breathing
- Administer 2/3 of that initial dose every hour
- Stop infusion if withdrawal signs or symptoms develop

NOTE: **If called upon to manage a naloxone infusion to treat severe opioid intoxication, defer to the sending or receiving physician with respect to dose and titration. In the absence of such guidance use the dosing in this profile.**

Pediatric

- Refer to adult dosing

Naloxone—Newborn Respiratory Depression

Neonatal

- 0.1 mg/kg IV/IM/IN

NOTE: **Respiratory depression refractory to PPV with a history of maternal opioid administration within 4 hours**

9.1.2 Cyanide Poisoning

Indications

- History of smoke inhalation or removal from fire along with any neurological, cardiac or respiratory complaint.

Hydroxocobalamin

Adult

- 5 gm IV infusion over 15 min
- May repeat 1 time
- Max total dose 10 gm

Pediatric

- 70 mg/kg IV infusion over 15 min
- Max single dose 5 gm
- Max total dose 10 gm

NOTE: **May give undiluted IV push in cardiac arrest**

NOTE: **History of smoke inhalation, removal from fire, Ingestion, inhalation, or dermal exposure to cyanide combined with neurological, cardiac or respiratory complaint.**

HydroxocobalaminAdvisories

- Administration of the second dose may range from 15 minutes to 2 hours based on patient condition
- The recommended diluent is 0.9% Sodium Chloride
- Use the diluent included in the kit or follow kit instructions if a diluent is not included
- Use a separate intravenous line for administration due to multiple incompatibilities

9.1.3 Tricyclic Antidepressant (TCA) Overdose

Treatment is indicated if any on the following symptoms occur:

- Cardiovascular instability
- Hypotension
- Rapid CNS deterioration (seizures, coma).
- Prolonged QT interval
- Widened QRS
- Ventricular dysrhythmias

NOTE: Arrhythmias secondary to TCA should be treated first with sodium bicarbonate

Sodium Bicarbonate

Adult

- 50–100 mEq IV over 5–10 min

Pediatric

- 1–2 mEq /kg IV over 5–10 min
- Max single dose 100 mEq

NOTE: May give 1 dose—additional doses require a physician order

Cross reference		
Clinical Care Guidelines	23	7-1 - Neurocritical Care

9.2 Envenomation

- Fully expose and examine the site of the bite or sting
- Observe for edema, redness, and number of puncture sites
- Mark border of swelling with ink and time
- If practicable, immobilize affected extremities in a position that is lower than the patient's heart

⚠ CAUTION **The insect/arachnid/snake should not be brought along on the transport.**

9.2.1 Snake Bite (Pit Viper)

- Measure the circumference of the bitten extremity just proximal to envenomation site every 15 minutes.
- Administer CroFab (Crotalidae Polyvalent Immune Fab [Ovine])

Crotalidae Polyvalent Immune Fab

Adult

- 4–6 reconstituted vials in 250 mL of 0.9% NS IV
- 25–50 mL/hr for 10 minutes observing for allergic reaction
- If no reaction occurs, increase rate to 250 mL/hr

Ped

- No PHI Standing Orders or Indications

Scorpion Sting

Observe for signs of envenomation neurotoxicity

- Flushing
- Diaphoresis
- Agitation
- Salivation
- Roving eye movements (Opsoclonus)
- Limb jerking (these are myoclonic jerking NOT seizures)

Treatment

Secretions

- Manage secretions using suction
- Atropine may be given onetime for hypersalivation

⚠ CAUTION **In addition to standard indications, early advanced airway management may be needed if secretions cannot be managed with suction or atropine.**

Atropine

Adult

- 1 mg IV over 1 min, IM

NOTE: **May be given one time**

Pediatric

- 0.02 mg/kg IV 1 min, IM
- Max single dose 0.5 mg

Sedation and Clonus

- Fentanyl (preferred) or midazolam may be given for clonus or sedation.

NOTE: **Avoid long-acting opioids or benzodiazepines such as hydromorphone, morphine lorazepam or diazepam**
NOTE: **Avoid anticholinergics such as diphenhydramine**

Fentanyl

Adult

- 0.5–2 mcg/kg IBW IV over 1 min, IM, NAS
- Repeat every 10 minutes, PRN

Pediatric

- Refer to adult dosing

Midazolam

Adult

- 0.1 mg/kg IV
- Max single dose 5 mg

Alternately

- 0.2 mg/kg NAS, IM
- Max single dose 10 mg

NOTE: **Preferred benzodiazapine if IM or NAS administration is indicated.**
NOTE: **Repeat every 2–3 minutes until the seizure is controlled.**

Pediatric

- Refer to adult dosing